

COMP 579 Project Proposal

Proposed Topic: Distributional Reinforcement Learning: Reproducing and Evaluating the C51 Algorithm

Group 10

Background and Motivation

In traditional reinforcement learning, the expected reward of a given state-action pair is typically computed by value functions like $Q(s, a)$. However, this approach ignores the full distribution of possible returns, which may contain useful information about uncertainty and risk. Bellemare et al. (2017) introduced Distributional Reinforcement Learning, where the entire distribution of the reward, not just the expected value, is learned by the reinforcement learning model. They implemented this idea in the C51 algorithm, which was proven to have better performance than the traditional DQN algorithm when tested on Atari games. In this project, we will attempt to recreate the concept of distributional reinforcement learning and test if it has better performance than traditional reinforcement learning on smaller environments.

Methods

We will implement the C51 (Categorical DQN) algorithm proposed by Bellemare et al. (2017) and compare it with a standard Deep Q-Network (DQN) baseline. Due to computational limitations, experiments will be conducted on lightweight environments such as LunarLander, MountainCar, and/or MinAtar. To study the effect of modeling value distributions, we will perform an ablation study on the number of atoms used in the categorical distribution. We will analyze how the number of atoms affects learning stability, convergence speed, and final policy performance. All models will use the same neural network architecture and hyperparameters, and results will be averaged across multiple random seeds to ensure fair comparison.

References

Bellemare, M. G., Dabney, W., & Munos, R. (2017).
A Distributional Perspective on Reinforcement Learning. ICML.